
Beyond IID: Controlled Studies of Visual Cues, Domain Alignment, and Open-Set Rejection

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Abstract

This report examines four ways in which a model can be accurate on its training distribution and still fail in use, studied under one shared protocol. Controlled image interventions on three frozen backbones show that removing or corrupting colour is almost free, that moving the object barely matters, but that scrambling the image's global arrangement is costly, and that the models divide on this by what they were pretrained on rather than by their architecture. On PACS, where a model trained on photographs, paintings and cartoons is asked to recognise sketches, no adaptation method improves on simply training with the labelled data available. The two adversarial methods leave the domains as easy to tell apart as before, so their losses cannot be explained by having removed domain information; a probe applied to the same frozen features recovers most of the class structure, which places the failure in the classifier rather than in what it reads. When the target domain is withheld entirely, both domain-generalization methods improve on the same baseline, so access to unlabelled target data was not what decided the outcome. In open-set recognition, a classifier that is 94.9% accurate on its known classes still accepts roughly two thirds of unfamiliar images drawn from semantically similar categories, and stronger augmentation improves rejection without improving accuracy. A theme runs through all four: a measurement taken at one level of a model rarely licenses a claim about another. Code, configurations and machine-readable results: <https://github.com/mujeeb-asad/ATML-PA1>.

1 Introduction

Supervised learning assumes that training and test data share a distribution and a label set. Deployment honours neither. An object may appear in an unfamiliar rendering style, a test image may come from a domain never seen in training, or an input may belong to no training class at all. Accuracy in-distribution says nothing about *which* evidence produced it.

Task 1 alters one visual property at a time and asks which cues a model relies on, and whether a change in prediction is matched by a change in the feature. Task 2 supplies unlabelled target images and asks whether aligning source and target representations helps. Task 3 removes the target domain entirely and compares aligning the observed domains against seeking a locally stable solution. Task 4 asks when a classifier should decline to answer. Section 2 gives the shared protocol and the departures from it, Section 7 draws the results together, and Section 8 states what a single run of each experiment permits.

2 Experimental Setup

Every random choice uses seed 6304, and each experiment ran once, so no number below carries an error bar. Checkpoints are the best epoch under a fixed rule: mean macro-F1 across the three source validation sets for Tasks 2 and 3, CIFAR-10 validation accuracy for Task 4.

Datasets and models. Task 1 uses STL-10 with a stratified 80/20 training split and a class-balanced 500-image test subset. ResNet-50, ViT-B/16 and OpenCLIP ViT-B/32 are frozen and used only to produce features, each read by a linear head trained on cached features; zero-shot CLIP, which classifies by comparing an image against the prompt *a photo of a {class}*, is a fourth readout of the CLIP embedding. Interventions are applied to one 224×224 canvas before each model’s normalisation, so all four see identical pixels. Tasks 2 and 3 use PACS, whose seven classes appear as photographs, paintings, cartoons and sketches; the first three are labelled sources and Sketch is the target. Both share one protocol: identical per-domain 80/20 splits loaded from a saved index file, a ResNet-18 with a fresh seven-class head, the same crops and optimiser, and batches of eight images per source domain, with Task 2 adding 24 unlabelled target images per step. BatchNorm running statistics stay frozen at pretrained values, as required, so no method gains an unintended adaptation by letting them drift toward the source–target mixture. Task 3 loads Task 2’s source-only checkpoint as ERM, and the two agree to 0.0000 points. Task 4 treats the ten CIFAR-10 classes as known and sixteen CIFAR-100 test classes as unknown, eight close to a known class and eight further away, 800 images each; all models share one CIFAR-appropriate ResNet-18, and one cached set of logits and features per model feeds every score.

Protocol enforcement. The specification forbids certain data from reaching certain stages: no target labels in Task 2’s training or selection, no Sketch image at all in Task 3 before final evaluation, no CIFAR-100 image in Task 4’s training or thresholds. Rather than rely on care, the code makes a violation fail outright. Task 2’s target dataset is built without labels, Task 3 replaces every sketch image’s bytes with None until the final cell, and Task 4 downloads only the CIFAR-100 test split. Qualitative examples are drawn by seeded shuffle. Runs executed on one NVIDIA RTX PRO 6000 under torch 2.11.0+cu128; cudnn.deterministic was not set, so repeated runs are not bitwise identical.

Deviations. Five decisions depart from the specification, three of them stabilisation measures the course authorised class-wide provided they are documented. **(1)** Gradient clipping at norm 1.0 was added to every Task 2 and 3 run after a source-side instability; the threshold is arbitrary and the fraction of steps it binds on is logged. **(2)** DANN’s and CDAN’s discriminator reads an L2-normalised feature rather than the same 512-dimensional feature the classifier uses, because the domain loss is unbounded above and a backbone can reduce it by enlarging features instead of rotating them, which frozen BatchNorm cannot counteract: scaling a fixed batch by 1000 raises the raw domain loss from 0.70 to 145.8 and leaves the normalised version unchanged. The classifier still reads the unnormalised feature and DAN is unaffected; Section 4 returns to the consequence. **(3)** A feature scaler precedes the separability probe, fitted on the training portion only, because the specified logistic regression converges poorly on raw post-ReLU features. **(4)** An additional DANN run at domain-loss weight 0.1 was trained with the instructor’s approval, the main comparison keeping unit weight, and **(5)** two bounded studies were therefore run where one is asked: the λ_{MMD} sweep is the designated study and the domain-weight arm a diagnostic.

Choices left open. Not departures, but each affects a number. The MMD estimator excludes self-comparisons, since the specification fixes the kernel but not the estimator and the biased form reads 0.473 at eight samples per side on two samples from the *same* distribution, larger in Task 3 than the quantity measured; small negative values are therefore expected. CIFAR is read from pinned HuggingFace mirrors for download speed, so the split remains stratified but its membership differs from a torchvision run. PROSER’s detection score fixes at zero the bias the reference sweeps, the operating point being set by a different rule. Two descriptive analyses were added after the target lock, and one extra row appears in Task 1’s stability table. Values fixed by neither specification nor principle: AdaIN $\alpha = 0.8$, the 0.35 edge-correlation floor, 24 conflicts per class pair, 50 PCA components, 16 px as the stability displacement, and the clip norm.

3 Task 1: Inductive Biases and Feature Representations

Three frozen backbones are compared: ResNet-50, ViT-B/16 and CLIP ViT-B/32, each read by a linear head, with zero-shot CLIP a fourth reading of the CLIP embedding. All four land within 0.8 points on unaltered images, so what distinguishes them is how far each falls when one visual property changes. That ceiling sets the scale: at 97.6% only twelve of 500 images are wrong, so two points means ten more errors.

Colour matters a little, and the wrong colour matters more. Grayscale costs every model about two points, and the 0.8-point spread between them is four images. Rotating the hue by half a turn costs more than removing colour altogether, for all four readouts: grayscale withholds a cue, a rotated hue supplies one pointing the wrong way. Only ResNet’s gap exceeds noise, at 4.4 against 2.0 points, making it the one model genuinely misled by colour.

Condition	Model	Top-1 (%)	Δ Top-1	Macro-F1	Consistency (%)
clean	resnet50 head	97.60	0.00	97.61	–
clean	vit_b_16 head	97.40	0.00	97.39	–
clean	clip head	97.60	0.00	97.59	–
clean	clip zero shot	96.80	0.00	96.75	–
grayscale	resnet50 head	95.60	-2.00	95.59	96.40
grayscale	vit_b_16 head	95.60	-1.80	95.59	95.20
grayscale	clip head	95.60	-2.00	95.57	95.60
grayscale	clip zero shot	94.20	-2.60	94.11	95.20
hue rotated	resnet50 head	93.20	-4.40	93.25	94.60
hue rotated	vit_b_16 head	94.20	-3.20	94.18	94.60
hue rotated	clip head	95.00	-2.60	94.94	95.60
hue rotated	clip zero shot	93.40	-3.40	93.31	94.20
patch shuffled	resnet50 head	89.80	-7.80	89.81	92.00
patch shuffled	vit_b_16 head	89.00	-8.40	88.89	90.00
patch shuffled	clip head	79.00	-18.60	78.94	79.00
patch shuffled	clip zero shot	78.80	-18.00	78.86	80.20

Table 1: Task 1 clean, grayscale, hue-rotated and patch-shuffled performance on the same 500 images. Δ Top-1 is against each model’s own clean baseline; consistency is the share of predictions unchanged. Confidence is comparable within a model, not across.

3.1 Shape versus texture

Cue-conflict images, built with AdaIN style transfer [Huang and Belongie, 2017], carry the shape of one class and the texture of another; a prediction naming neither counts toward coverage but not toward shape bias, following Geirhos et al. [2019]. Every model prefers shape, ordered ResNet < ViT < CLIP. ResNet reaches 67.7% and ViT 83.2%; their confidence intervals overlap by 0.3 points, but interval overlap is conservative for two proportions, and a direct test on 65/96 against 109/131 gives $z = 2.73$, $p = 0.006$.

Coverage limits what the figure can carry. ResNet resolves only 47.1% of conflicts, so its score describes 96 of 204 images and the remaining 108 leave its shape share anywhere between 31.9% and 84.8%. That is not merely a small denominator, since it also makes the most texture decisions outright: 31, against 22, 5 and 6. CLIP’s 96% rests on five such decisions, so one flipped image moves it 0.8 points. Across both experiments ResNet is the model most misled by colour and the least shape-biased, CLIP the most shape-dominant and least colour-sensitive; the two CLIP rows differ by two or three decisions, placing this preference in the representation rather than in what reads it.

3.2 Translation and patch structure

Shifting the object produces almost nothing: the largest accuracy change anywhere is 1.2 points, several entries at 8 and 16 px sit *above* the unshifted baseline, and the four directional means span only 96.95 to 97.17. The experiment cannot resolve a positional effect and is reported as a null result.

Model	Shape	Texture	Other	Shape bias (%)	Coverage (%)
resnet50 head	65	31	108	67.71	47.06
vit_b_16 head	109	22	73	83.21	64.22
clip head	121	5	78	96.03	61.76
clip zero shot	119	6	79	95.20	61.27

Table 2: Cue-conflict decisions over 204 conflicts. Shape bias divides by decided images only; coverage gives the share decided. Raw counts are shown because the denominators differ sharply.

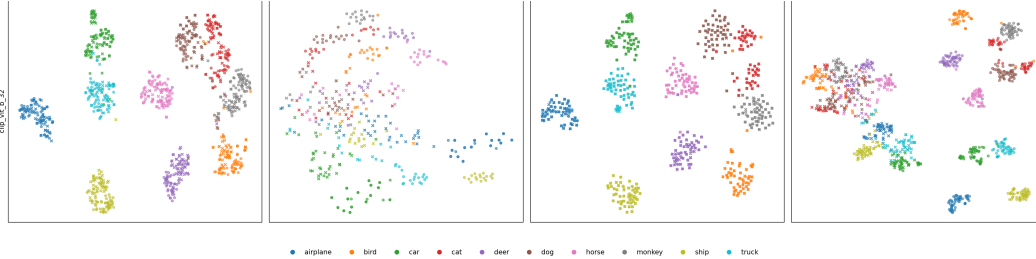


Figure 1: CLIP ViT-B/32 features, clean (circles) against transformed (crosses), coloured by ground-truth class. One t-SNE is fitted per panel, so coordinates cannot be compared between panels. ResNet-50 and ViT-B/16 panels are in the repository.

Prediction consistency, the share of predictions surviving the shift regardless of correctness, does fall steadily; at 32 px ResNet leads at 98.65% against CLIP zero-shot’s 97.30%, a spread of 0.9 points too small to support a claim. ViT’s consistency is non-monotonic (98.35, 98.90, 97.75%), and 16 px is exactly one ViT-B/16 token, so a shift of that size leaves token contents intact and changes only their positional embeddings where 8 px cuts through every token, which is what learned absolute positions would predict, on about eleven predictions.

Scrambling the image is a different story. A 4×4 shuffle of 56 px patches costs ResNet 7.8 and ViT 8.4 points but CLIP 18.6 and 18.0. **The division is not convolutional against transformer, since ResNet and ViT differ by three images, but CLIP against the two models pretrained on ImageNet.** Patch size does not explain it: a 56 px block spans about 3.5 ViT-B/16 tokens but only 1.75 CLIP tokens, so CLIP’s coarser tokens should survive better and instead it loses twice as much. Confidence agrees. ResNet falls only from 0.895 to 0.869 and ViT from 0.962 to 0.900 while both lose eight points, so both still find evidence they trust in the surviving patches; CLIP zero-shot falls from 0.943 to 0.775, in step with its accuracy.

Backbone	cue conflict	grayscale	hue rotated	patch shuffled	translation 16px
clip_vit_b_32	0.680	0.893	0.894	0.738	0.968
resnet50	0.276	0.788	0.823	0.718	0.952
vit_b_16	0.258	0.739	0.689	0.652	0.974

Table 3: Mean cosine similarity between each clean feature and its transformed counterpart, paired per image. Comparable within a backbone, not across.

3.3 Representation stability

Cosine stability asks how far the feature itself moves, and disagrees with the predictions twice in opposite directions. ResNet’s hue rotation moves its features *less* than grayscale (0.823 against 0.788) yet flips more predictions (27 against 18) and costs 2.2 times the accuracy, so the movement lies along directions the classifier reads and its size is a poor guide to its consequences. Running the other way, CLIP has the *highest* cosine stability under patch shuffling (0.738, against 0.718 and 0.652) while losing by far the most accuracy. These values cannot be calibrated, since no random-pair baseline was computed and the three feature spaces are not comparable, but Figure 1 shows what a single number cannot: CLIP’s shuffled features leave their clean clusters for a class-mixed region, animals with

animals and vehicles with vehicles. The shift is small in angle and still crosses class boundaries. Cue conflict gives the lowest cosine for every backbone (0.28, 0.26, 0.68), but that figure is confounded, since AdaIN re-synthesises the whole image and no $\alpha = 0$ reconstruction was measured.

CLIP’s two readouts share one frozen embedding, so their cosine stability is identical and any behavioural difference is the readout alone: across every intervention their accuracy changes differ by at most 0.8 points, their patch collapse is equal, and their shape bias differs by one point. **Here it is the embedding, not the readout, that determines the decision.**

What can be attributed to architecture. ResNet-50 and ViT-B/16 share a corpus and objective, differing in architecture, capacity (25.6M against 86.6M) and augmentation recipe, which makes them the cleanest pair available; their shape-bias difference is the only effect attributed to architecture here, and capacity and recipe still differ. ViT-B/16 against CLIP ViT-B/32 varies patch size, corpus, objective, embedding width and normalisation at once, so CLIP’s patch collapse and 96% shape bias are not attributed to architecture: being a transformer is clearly insufficient, since ViT-B/16 loses only 8.4 points on the same images. Two limits bear on the CLIP rows: the backbone was built without QuickGELU, which is not the activation its weights were trained with, so every CLIP number carries an unquantified offset, and per-image predictions were not retained, so which classes absorbed the undecided conflicts cannot be recovered. The weakest evidence here is the translation experiment, the second weakest any cross-backbone cosine comparison.

Method	Photo F1	Art F1	Cartoon F1	Mean src F1	Sketch acc.	Sketch F1	Δ Sketch acc.	Separability (%)
Source-only	95.61	90.45	95.05	93.70	65.18	64.15	0.00	99.86
DAN	94.94	89.64	94.20	92.93	56.91	55.31	-8.27	84.34
DANN	94.25	87.21	94.94	92.13	31.81	31.52	-33.37	99.86
CDAN	95.17	88.10	93.60	92.29	14.58	24.09	-50.60	100.00

Table 4: Task 2 main comparison. Separability is a two-way probe on source-validation against target features, chance 50%; source-only already reaches 99.86%.

4 Task 2: Unsupervised Domain Adaptation

A ResNet-18 is trained on photographs, paintings and cartoons and evaluated on sketches, visible during adaptation as unlabelled images and labelled only once every checkpoint and setting is fixed. Four methods (source-only ERM, DAN, DANN and CDAN) share one training loop, initialisation and sequence of source batches, so the alignment term is the only difference. Two probes of identical construction accompany the accuracy figures, one predicting the domain a feature came from and one predicting its class. **No method beats source-only ERM at its specified setting** (Table 4), and the three failures have two different causes.

How large the gap is, and where it falls. Source-only reaches 93.76% on source validation and 65.18% on sketches, **a gap of 28.6 points**, with Art the weakest source for every method. The failures avoid the rare classes: house and person, at 80 and 160 sketch images, score 76.3 and 84.4%, while the worst are dog at 23.8% and giraffe at 37.7%, both among the largest. Every dominant confusion is one four-legged animal for another (giraffe for horse, 38.1%; dog for horse, 37.0%; dog for elephant, 19.8%), and horse absorbs 35% of all predictions against a true share of 20.8%, so even its 88.7% recall is partly a by-product of acting as a catch-all. With shading and colour gone only silhouettes remain, and separating four-legged animals from silhouette alone needs finer detail than the model learned; guitar, whose outline is distinctive, holds at 94.7%.

Two failure modes, and for two methods no alignment at all. Across seven runs the correlation between separability and target accuracy is $\rho = -0.58$ ($p = 0.18$), which establishes nothing. The prior observation matters more: source-only already scores 99.86% on the separability probe, one misclassified image in 728, so the domains were never hard to tell apart and any method that aligned them should have moved that toward 50%. **Only DAN moved it**, to 84.34%; DANN’s is 99.86%, identical to the unadapted baseline, and CDAN’s 100%. Whatever cost those two their 33 and 51 points, it was not the removal of domain information. DAN’s damage instead follows the label-shift pattern below, while DANN and CDAN collapsed onto a single class, DANN assigning 57.7% of all sketch predictions to house and CDAN 89.3% to person, sending 96.8 to 99.5% of horse, dog

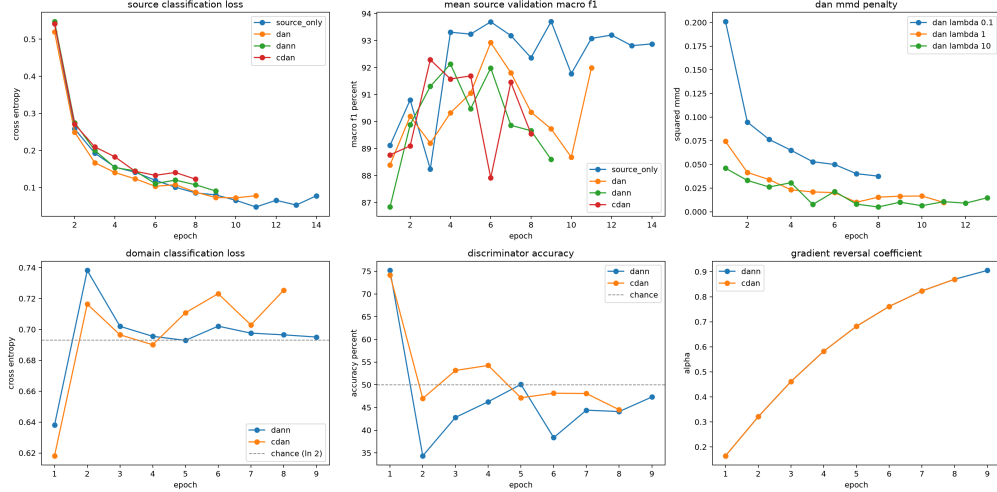


Figure 2: Task 2 training curves: source classification loss, mean source-validation macro-F1, the DAN MMD penalty at each λ_{MMD} , and the domain loss, discriminator accuracy and reversal coefficient for DANN and CDAN against their chance levels.

and elephant to that one label, all with source validation intact at 92.1 and 92.5%. Negative transfer appears at both levels: DAN loses 842 images source-only classified correctly and gains 517, DANN loses 1,491 and gains 180, CDAN loses 2,022 and gains 34, with elephant, horse and guitar damaged by all three and no class helped by all three.

Why the discriminator and the probe disagree. DANN’s domain loss sits at $\ln 2$ from the third epoch, its discriminator averages 43.5% accuracy, and its largest domain logit never exceeds 0.48: by its own account the domains are indistinguishable. Deviation (2) explains why that does not contradict the probe. The discriminator reads a normalised feature and can only constrain *direction*; the probe reads the raw feature, whose average magnitude grows from 44 to 108 over training. **The probe uses exactly the information the normalised domain loss cannot touch**, so a discriminator at chance is evidence about directions, not about domain information in general.

The class information survived; the classifier did not. The class probe recovers between 83.5 and 88.7% on sketches in *every* run, against 88.6% for source-only and 92 to 94.5% on the source side, so the distance between what the features support and what the model delivers is 31.3 points for DAN, 51.6 for DANN and 70.0 for CDAN. A genuine decline of four to five points appears for DANN, CDAN and the strongest DAN setting, but nothing justifying the claim that class structure was destroyed. **The familiar warning that alignment strips out class-discriminative information does not describe what happened here:** the adversarial objective damaged the readout on sketches without making the domains harder to distinguish. The probe is trained on sketch labels, so it bounds what a linear model could extract rather than promising a label-free method could find it.

4.1 Class-conditional alignment

CDAN is worse than both marginal methods on every target metric, at 14.58% accuracy and 24.09 macro-F1, yet its class probe is within a point of DANN’s, so the difference lies in the readout. CDAN hands the discriminator the outer product of feature and predicted class probabilities, placing the feature in the block of the predicted class; once the classifier collapses, almost every target image occupies the same block and the discriminator can separate the domains by noticing which block is filled, consistent with its perfect separability and with domain logits reaching 0.6 to 1.3 where DANN’s stay below 0.48. Neither factor is detached, so the reversed gradient also reaches the classifier head. Per-epoch target entropy was not logged, so this follows from the code and recorded prediction distributions rather than from measurement. **The result cannot settle whether conditioning preserves semantic structure:** conditioning on a collapsed prediction is uninformative by construction, and the collapse was already present at the selected checkpoint, epoch 3. CDAN

is also the only run whose macro-F1 exceeds its accuracy, by 9.5 points, because person reaches 100% recall on its 160 images and house 63.8% on its 80 while the five large classes sit near zero, a reminder that macro-F1 can flatter a collapsed predictor.

λ_{MMD}	Mean src acc.	Mean src F1	Separability	Target acc.	Target F1	Δ target acc.	Src-val pick
0.10	92.57	92.91	98.76	75.49	71.75	10.31	False
1.00	93.21	92.93	84.34	56.91	55.31	-8.27	True
10.00	87.93	87.82	85.03	68.44	67.28	3.26	False

Table 5: DAN at $\lambda_{\text{MMD}} \in \{0.1, 1, 10\}$. Source-validation selection chooses $\lambda = 1$ by 0.019 macro-F1, the worst of the three on target.

4.2 Alignment strength

The expectation recorded before this sweep was read was that separability would fall steadily, source performance would hold then slip at the strongest setting, and target accuracy would peak in the middle. Only the middle prediction held: source macro-F1 is flat at $\lambda = 0.1$ and 1 and falls 5.1 points at 10. Separability drops once and stays flat, with $\lambda = 10$ five probe images *above* $\lambda = 1$, and target accuracy forms a V whose minimum is the specified setting: 75.49, 56.91 and 68.44%. **The best target result in the study came from the weakest alignment**, at 98.76% separability. Nothing establishes why the response is non-monotonic; seed variance, the different selected epochs (3, 6 and 8) and the label-shift pull at $\lambda = 1$ remain candidates. At $\lambda = 10$ the penalty crowds out classification: the loss starts at 1.05 and settles at 0.32 to 0.58 against 0.05 to 0.10 elsewhere, gradient norms average 32 to 59 with a peak of 425, clipping binds on every step, the mean feature norm reaches 468, and the class probe drops 4.4 points.

Choosing without target labels means choosing on source validation, which picks $\lambda = 1$ by 0.019 macro-F1, roughly seventy times smaller than the run’s own epoch-to-epoch variation and twenty-five times smaller than the run-to-run noise of Section 8. It selects the worst of the three at a cost of 18.6 points. Source validation can reject $\lambda = 10$, which loses five points on the source side, but cannot separate 0.1 from 1, which is exactly where the target difference lies.

A limit the data imposes on any method. Person makes up 21.7% of the source mixture and house 14.5%, against 4.1% and 2.0% in sketches; the total variation between the two class distributions is 0.3007. Perfect alignment would make the predicted-label distributions identical, and the triangle inequality then gives $\varepsilon_T \geq \text{TV} - \varepsilon_S$ [Zhao et al., 2019]: source error can be traded for target error, but their sum cannot fall below the mismatch. With source-only’s 6.2% source error that caps sketch accuracy at 76.2%, so DAN at $\lambda = 0.1$ reaching 75.49 against its own 77.4% ceiling is consistent rather than paradoxical; the 69.9% figure sometimes quoted assumes no source error and applies to none of these models. The bound reads target labels and was computed after the lock.

It accounts for part of the outcome. DAN’s predicted distribution moved toward the source prior, from a total variation of 37.6 to 11.7, and its losses fall where the mechanism says: horse −290 images, elephant −253, guitar −85, all under-represented in that prior. **But its two largest gains, dog at +198 and giraffe at +94, are under-represented as well**, so the mechanism does not cover them; those come from horse absorbing far fewer images, its share falling from 34.8% to 13.0%. It does not apply to DANN or CDAN, which overshoot the prior rather than converging on it and reduced no separability.

The domain-weight arm. Lowering DANN’s domain-loss weight from 1 to 0.1, trained with the instructor’s approval as a stability diagnostic, lifts sketch accuracy from 31.81 to 67.80%, a 36-point swing that also puts it 2.6 points above source-only. Neither weight aligned the domains (99.86 and 99.73%), so the improvement comes from the adversarial signal interfering less with the classifier rather than from better alignment. The finding concerns this configuration, not the method.

5 Task 3: Domain Generalization

Sketch is now withheld completely: no sketch image reaches training, the source-side diagnostics, checkpoint selection or any hyperparameter choice. ERM is Task 2’s source-only checkpoint loaded

unchanged, so the three methods differ only in what they optimise. DAN-DG applies the same discrepancy measure between the three pairs of *observed* source domains rather than between source and target. SAM removes no domain information at all, instead seeking a solution whose loss changes little under a small weight perturbation. Two diagnostics run before the lock: how well a classifier identifies which source domain a feature came from, and how much the loss rises after one small ascent step.

Both methods beat the shared baseline. DAN-DG reaches 71.93% on sketches and SAM 74.06%, 6.74 and 8.88 points above ERM against a sampling standard error of about 0.7. **This reverses Task 2, where every method had strictly more information and every method lost ground.**

Source performance identifies the best model but cannot separate the other two. SAM leads on mean source accuracy (96.19), on its weakest domain (93.66, Art in every case) and on sketches. Between ERM and DAN-DG the source numbers are useless: ERM is ahead by 0.2 points on the mean and 0.5 on the worst domain, both inside the noise band, while DAN-DG is 6.7 ahead where it counts. Source metrics fail to resolve a real difference rather than pointing the wrong way, and the worst-domain figure adds nothing. Class-level failures stay hidden in both directions: ERM scores 23.8% on sketched dogs despite a source macro-F1 of 93.7, and DAN-DG improves overall while person collapses by 43.75 points, 70 of its 160 images with 48% predicted giraffe, its source macro-F1 down only 0.39.

DAN-DG is the one case where invariance and performance move together. Source-domain separability falls from 84.72% to 68.44%, closing 31.7% of the distance to chance against a probe error of about 2.1 points, with the source side unharmed. Two things separate it from Task 2’s DAN: it aligns *labelled* domains, so every feature being pulled together also carries a class loss holding it in place, and the class distributions being aligned differ far less, a total variation of 0.11 to 0.18 against 0.30. The single casualty fits that remaining mismatch, since person is both the class whose frequency varies most across the sources (25.9, 21.9, 17.3%) and the one most over-represented relative to sketches. The data are consistent with this; it is not established.

λ_{DG}	Mean src acc.	Mean src F1	Worst src acc.	Separability	Sketch acc.	Sketch F1	Δ Sketch acc.
0.10	93.79	93.92	90.98	83.39	64.55	65.80	-0.64
1.00	93.55	93.31	90.00	68.44	71.93	71.92	6.74
10.00	32.94	28.30	22.44	48.50	35.30	27.59	-29.88

Table 6: DAN-DG at $\lambda_{DG} \in \{0.1, 1, 10\}$. At $\lambda_{DG} = 10$ the model never learns, so its low separability is degeneracy rather than invariance.

5.1 Penalty strength and local stability

At $\lambda_{DG} = 10$ the model never learns. Mean source accuracy is 32.9%, its weakest domain 22.4%, and the classification loss sits between 1.72 and 1.97 for all 21 epochs against a chance value of 1.95; in the first two epochs source accuracy matches what predicting one class for every image would score. The penalty reaches zero by the second epoch with 8 of 21 epoch means slightly negative, as an unbiased estimator near zero should be. **Its separability of 48.5%, the closest to chance anywhere in this study, is degeneracy rather than invariance.** Whether alignment removed class information or prevented it being learned cannot be decided here, since the loss was already at chance in the first epoch and Task 3 has no class probe.

SAM lowers the sharpness measure, but that measure is close to circular. The ordering is SAM 0.130, DAN-DG 0.428, ERM 0.459, matching the sketch ordering. Adding the $\lambda_{DG} = 0.1$ run breaks it, that model being second flattest at 0.223 and worst of the working models on sketches at 64.5%, which drops the correlation across four models to -0.40 . More fundamentally, the measure is the loss increase after one normalised ascent step of radius 0.05, and SAM trains to minimise that exact quantity at that exact radius, so its winning shows it optimised its own objective in a way that carries to held-out source images. Base losses differ threefold, and normalising by them keeps SAM flattest. At $\lambda_{DG} = 10$ the measure reads 40.3, so it is not meaningful away from a working solution.

Score	AUROC near	AUROC far	AUROC all	Known accept (%)	Near reject (%)	Far reject (%)
MSP	81.93	90.22	86.08	94.69	28.25	46.38
MLS	80.40	90.00	85.20	95.20	30.88	54.12
Energy	80.41	90.09	85.25	95.25	30.75	56.00
Mahalanobis	79.84	92.43	86.14	94.63	28.25	52.75

Table 7: Four post-hoc scores on the frozen vanilla model, all reading one cached set of logits and features. Rejection is measured at the validation-calibrated threshold.

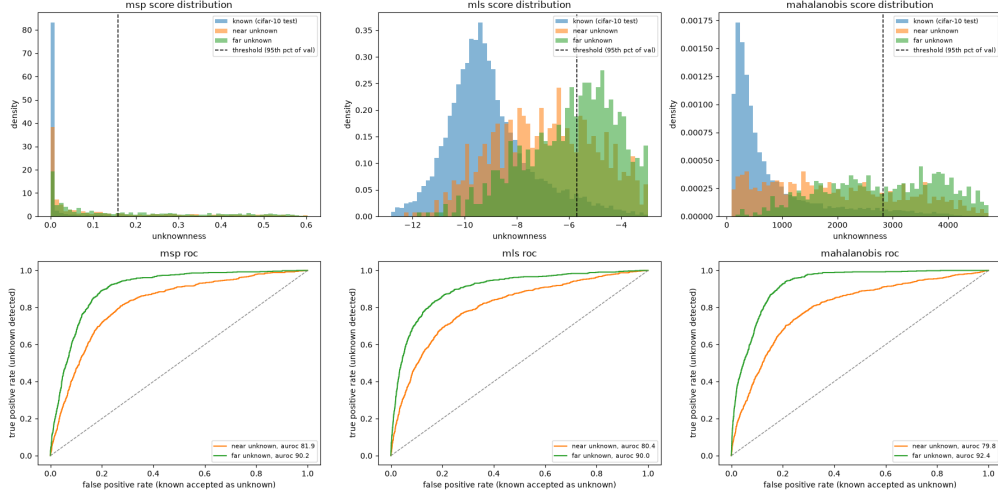


Figure 3: Score distributions for knowns against near and far unknowns, with the corresponding ROC curves, for three of the four scores on the frozen vanilla model.

SAM’s advantage cannot be cleanly credited to flatness. It improved the source side by 2.4 points, and its source-domain separability (85.71) is within probe error of ERM’s (84.72), so **the best domain-generalization model here changed domain invariance not at all**. The competing explanation is better optimisation: SAM runs two forward and backward passes per step and trained 20 epochs against ERM’s 14, and at the matched ninth epoch its source advantage is +0.24, which is noise. Telling the two apart would need a compute-matched ERM or several seeds.

What unlabelled target data contributed. Against the shared ERM baseline, DAN with unlabelled sketches reaches 56.91% and DAN-DG without them 71.93%: **the method that saw the target is 15.0 points worse than the one that did not, and worse than doing nothing**. Target access can still help, since Task 2’s DAN at $\lambda = 0.1$ gave the best sketch result in the study, but only at a setting label-free selection rejected. The comparison is not causal, because target access varies together with what is aligned, the samples per estimate (24 against 24, versus 8 against 8 per pair), the bandwidth pool, whether the aligned features carry a class loss, and how far apart the class distributions are. Held fixed are the optimiser, budget, initialisation, batch order, stopping length and the discrepancy code. The finding concerns this way of using unlabelled sketches, not unlabelled target data in general.

6 Task 4: Open-Set Recognition

The ten CIFAR-10 classes are treated as known and sixteen CIFAR-100 classes as unknown, eight close to a known class and eight further away, 800 images each. Four scores are computed on one frozen classifier from a single cached set of logits and features: maximum softmax probability, maximum logit, energy and Mahalanobis distance. Two further models ask whether the representation itself can improve: GCSC is the same recipe with stronger augmentation and nothing else changed, and PROSER fine-tunes the trained classifier with five extra dummy outputs and synthetic examples interpolated between known classes [Zhou et al., 2021]. Every threshold is the 95th percentile of unknownness on CIFAR-10 validation data, so roughly 95% of known images are accepted.

Group	Unknown class	Predicted known	Score	Threshold
near	wolf	dog	-7.567	-5.711
near	tractor	truck	-7.792	-5.711
near	pickup_truck	automobile	-6.777	-5.711
far	mushroom	cat	-11.068	-5.711
far	bowl	frog	-9.693	-5.711
far	wardrobe	truck	-5.839	-5.711

Table 8: Accepted unknowns under the vanilla MLS threshold. A score further below the threshold is a more confident acceptance.

Model	Score	CSA (%)	AUROC near	AUROC far	AUROC all	Near reject (%)	Far reject (%)
Vanilla	MLS	94.88	80.40	90.00	85.20	30.88	54.12
GCSC	MLS	95.01	80.62	91.39	86.01	32.75	59.38
PROSER	MLS	94.30	79.83	88.67	84.25	28.00	51.38
PROSER	Placeholder	94.30	77.87	89.07	83.47	25.62	51.00

Table 9: Vanilla, GCSC and PROSER under MLS, with PROSER’s placeholder score as an additional row. Closed-set accuracy uses the ten known logits throughout.

Semantic distance governs detectability, and the best score depends on which kind. Every score finds distant unknowns easier than close ones, by 8.3 to 12.6 AUROC points, and the ranking flips between them: maximum softmax probability is best on near at 81.9 and joint-worst on far, Mahalanobis worst on near at 79.8 and best on far at 92.4. Only those two leads exceed about one standard error. A distant unknown occupies a different region of feature space, which a distance measure sees, whereas a close unknown sits inside a known class’s region where distance says little. Maximum logit and energy differ by 0.02 and 0.08 points, since at 99.98% training accuracy the softmax is peaked enough that summing over logits reduces to taking the largest.

The operating point tells a harsher story. At 95% known acceptance the maximum-logit score accepts 69% of near unknowns, 553 of 800, as though they were known classes, and 45.9% of far ones. Maximum softmax probability, which wins on near AUROC, is joint-worst on near rejection and 7.8 points behind on far, because its validation scores bunch so close to zero that the 95th percentile has little to work with. Achieved acceptance varies between 94.63 and 95.25%, so the rates are not quite measured at the same point. Near unknowns are absorbed mainly by truck (155), cat (104) and automobile (99), far unknowns by cat (80), bird (74) and truck (66); the rankings barely agree ($\rho = 0.125$), though cat and truck are high in both. Within the near group 274 absorptions go to vehicles and 279 to animals, matching its even composition, while horse takes only 18 despite camel being an obvious neighbour.

Three of the six recorded failures need no explanation: a wolf called a dog, a tractor called a truck, a pickup truck called an automobile, each within 1.1 to 2.1 of the threshold. A wardrobe called a truck is visually reasonable and 0.13 inside the boundary, and a bowl called a frog is plausible on colour and texture if not on meaning. **A mushroom called a cat has no such account and is the most confident of the six**, 5.36 below the threshold, further inside than a typical known image, so no confidence threshold near this operating point could reject it without rejecting many knowns.

6.1 Augmentation and placeholders

Stronger augmentation improved rejection without improving classification. GCSC gains 0.13 points of closed-set accuracy, below the 0.22-point standard error. Its rejection improves: +0.23 AUROC on near unknowns, +1.39 on far, and 5.25 points of far rejection at unchanged known acceptance, about three standard errors and its clearest single effect. **The gain on far unknowns is roughly six times the gain on near ones.** Augmentation kept the training loss away from zero (0.032 against 0.0024 at the last epoch), which plausibly leaves the model less overconfident on inputs far from anything it saw while doing nothing for close ones, which still sit inside known regions; no logit statistics were saved, so that account is untested. Closed-set and open-set performance did not move together.

PROSER was worse on every measure, and the selection rule explains why. It gives up 0.58 points of closed-set accuracy, loses 0.57 and 1.33 AUROC points on near and far unknowns, and its own detection score is 1.96 points worse than plain maximum logit on near unknowns, the case it exists for. **The selected checkpoint is epoch 4 of 50**, where the loss term governing the synthetic examples still sits at 1.975 against a chance value of 2.40. The other placeholder term converged by epoch 2, but this one only begins training around epochs 6–7, exactly when validation accuracy starts to fall. Selecting on validation accuracy therefore picks a checkpoint from before the mechanism engaged, which follows from the rule rather than the implementation but means the behaviour PROSER was fine-tuned to produce was barely present in the model evaluated. PROSER also differs in having 50 extra epochs at a different learning rate, with no matched control.

Where interpolation runs out. The synthetic examples fill the space *between* known classes, but the failures sit *inside* them: the mushroom called a cat and the bowl called a frog both carry large logits and lie between no two CIFAR-10 classes, and even the wolf called a dog sits within the dog region. Interpolating between known classes cannot produce a stand-in for an unknown arriving from a direction no pair of known classes spans.

7 Cross-Task Discussion

Visual cues. Task 1 would predict a small PACS gap, since colour costs its backbones about two points, and the real gap is 28.6. No Task 1 intervention reproduced what a sketch does: grayscale retains shading and local texture, which a line drawing discards, while global arrangement, the fragility Task 1 found most severe, is what sketches preserve. What survives the comparison is narrower. Once surface evidence is gone a coarse silhouette still separates guitar, house and person but not four-legged animals, which is why every PACS model’s dominant confusions are between quadrupeds. Datasets, backbones and training regimes all differ, so this is qualitative.

Invariance against discriminability. Pairing separability with target accuracy across both tasks, read separately since the probes have different chance levels, yields no consistent relationship. One run has invariance rising with target performance while the source side holds, DAN-DG at $\lambda_{DG} = 1$, and even it cost person 43.75 points. Two have target performance collapsing while separability does not move, where the usual diagnosis is simply wrong: no invariance was achieved, so none can have been traded away. The clearest case of the failure the assignment warns about, $\lambda_{DG} = 10$, is a model that never learned. **Across seven alignment runs invariance was neither necessary nor sufficient:** the best model in each task changed separability barely at all, and the most invariant models were among the worst. Target access is the same story in miniature, and Section 5 gives it: the method that saw unlabelled sketches finished 15 points below the one that did not.

Recognition against rejection. Every Task 4 closed-set difference is at most 0.6 points against a standard error of 0.22, so the three models cannot establish that accuracy and rejection move together. GCSC settles it the other way, gaining five points of far rejection at flat accuracy. The same split between what a representation holds and what a decision rule extracts runs through Task 2, where class structure survived classifiers that had collapsed, and Task 1, where CLIP’s two readouts of one frozen embedding behave almost identically while its features move further under patch shuffling than ResNet’s.

Synthesis. A representation should keep the class structure that survives a change of surface, and here it largely did: Task 2’s probe recovered 83 to 89% of sketch classes from features whose classifiers delivered 15 to 32%. What it should discard is equally clear, but *forcing* that invariance did not produce it. Adversarial forcing never achieved it, strong marginal forcing prevented learning altogether, and the two methods that did improve on an unseen domain either aligned already-labelled domains or changed invariance not at all. Rejection of inputs outside the label space was unreliable for every method tested, worst for the near cases most likely to arrive, and occasionally failing with more confidence than a genuine known attracts. Of the seven prescribed methods, two clearly beat their baseline, one marginally, and four lost.

8 Limitations

Each experiment ran once, so **no result carries an error bar**, and with cuDNN left non-deterministic repeated runs are not identical: Task 2’s source-only run, executing byte-identical code, trained 23 epochs on one occasion and 14 on another, mean source macro-F1 differing by about half a point. **Differences below roughly one point should not be read as effects**, which rules out Task 1’s translation result, GCSC’s accuracy change, and most near-unknown differences between scores. Three deviations changed what the methods optimise, so every conclusion about DAN, DANN, CDAN and DAN-DG concerns the modified objective: gradient clipping, which binds on 77 to 100% of steps and is therefore part of the optimiser in every run including the baseline; the choice of MMD estimator; and the normalised discriminator input, whose interaction with a probe reading raw features is analysed in Section 4. The class probe is trained on target labels and reports an upper bound on what is linearly recoverable. Task 1’s CLIP backbone was built without QuickGELU, so every CLIP number carries an unquantified offset, and per-image predictions were not kept. Target labels were read once, on the selected checkpoint, so the gap between it and the best-transferring epoch cannot be measured. Several mechanisms proposed above (the non-monotonic sweep, CDAN’s degenerate conditioning, why cat and truck absorb so many unknowns) are consistent with the recorded diagnostics but were not measured directly.

9 Conclusion

The methods that improved on their own baselines strengthened or regularised the supervised solution: aligning source domains that were already labelled, seeking a locally stable solution, and augmenting more aggressively. Those that tried to impose a property on the representation, by adversarial training or strong marginal alignment, did not, and in two cases damaged the classifier without achieving the property at all. The diagnostics are what make this more than an impression: a separability probe showing that alignment never occurred, a class probe locating the failure in the readout rather than the features, and a label-shift bound showing what the data makes impossible for any method. A measurement at one level licenses no claim at another. A discriminator at chance is not an aligned representation, low domain separability is not preserved class structure, and a 94.9% accurate classifier is not a safe one.

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